

CouplingNet II: Branches and Local Context for Split Prediction

Branch nets encode profiles; trunk nets encode pointwise coordinates

BRANCH NETS

SOURCE PROFILE

sample-varying forcing profile

COEFFICIENT PROFILES

diffusion, primary/transverse convection, reaction

LINE-GEOMETRY STRUCTURE

interval length, endpoints, transverse placement

SPLIT PREDICTOR

(branch features, trunk features) $\rightarrow (\phi, \psi)$

TRUNK NETS

AXIAL LOCAL TRUNK

$t_{\parallel}$: position along the primary interval

POINTWISE TRANSVERSE TRUNK

$t_{\perp}$: position on the transverse interval through this point

$\underbrace{\text{profiles and line geometry}}_{\text{branch nets}} + \underbrace{\text{pointwise coordinates}}_{\text{trunk nets}} \Rightarrow \text{directional source split.}$